

Nmap Network Scan Demo (Basic Package)

This is a demo of my process for performing network scans using Nmap for clients purchasing the Basic Package. Clients can request documentation such as this if they would like to. For this Nmap network scan demo, I will be scanning my own local host, which possesses the ip address: 127.0.0.1.

1) Basic Port Scan (Quick Overview of Open Ports)

- Command: `nmap localhost`
- What it does: Performs a default TCP SYN scan on the top 1000 common ports. Identifies open, closed, or filtered ports. This is non-intrusive and fast.
- Run it: Type the command and press Enter.
- Expected time: 1-2 minutes.
- Ensure there are open ports for the sake of the demo:
 - I decided to start SSH service (Port 22)

- `sudo systemctl enable --now ssh`
- `sudo systemctl status ssh --no-pager`

```
(root@kali)-[/home/kali]
# sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

(root@kali)-[/home/kali]
# sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-10-16 22:29:00 EDT; 11s ago
     Invocation: d6e7633a61f7a18e9b50614dec75d139
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 10702 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 10704 (sshd)
      Tasks: 1 (limit: 4456)
     Memory: 2.1M (peak: 2.6M)
        CPU: 27ms
    CGroup: /system.slice/ssh.service
            └─10704 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 16 22:29:00 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 16 22:29:00 kali sshd[10704]: Server listening on 0.0.0.0 port 22.
Oct 16 22:29:00 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Oct 16 22:29:00 kali sshd[10704]: Server listening on :: port 22.
```

```
(root@kali)-[/home/kali]
# nmap 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-16 22:29 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

Interpretation: Port 22 (SSH) is open. 999 of the other most common ports are closed. Since port 22 (SSH) is the only one open, this is minimal exposure, which is good in terms of network security. SSH is common, though if not secured it can be vulnerable to brute force attacks.

2) Service Version Detection (Identifies Software on Open Ports)

- Command: `nmap -sV localhost`
- What it does: Builds on the basic scan by probing open ports for service names and versions (e.g., "Apache 2.4"). This helps identify potential vulnerabilities based on outdated software.
- Run it: If on Linux/macOS, use `sudo nmap -sV localhost` for better accuracy. On Windows, run Command Prompt as admin.
- Expected time: 2-5 minutes.

```
(root@kali)-[/home/kali]
# nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-16 22:35 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 8 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds
```

Interpretation: The target is running OpenSSH 10.0p2 (Debian 8). The latest version as of October 2025 is 10.2p1, which includes additional security fixes and stability improvements. Updating OpenSSH to the latest version is recommended.

3) Vulnerability Scan (Detects Known Vulnerabilities):

- Command: `sudo nmap --script vuln localhost` (use sudo on Linux/macOS; run as admin on Windows). Add -v for verbose output: `sudo nmap -v --script vuln localhost`.
- What it does: Uses Nmap's Scripting Engine (NSE) to run vulnerability scripts on open ports. It checks for common issues like outdated protocols or exploitable services. This is key for your gig's "vulnerabilities" listing.
- Prerequisites: Update scripts with `sudo nmap --script-updatedb` (one-time).
- Expected time: 5-15 minutes (scripts can be thorough).

```
(root@kali)-[/home/kali]
# sudo nmap -v --script vuln 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-16 22:39 EDT
NSE: Loaded 105 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 22:39
Completed NSE at 22:39, 10.01s elapsed
Initiating NSE at 22:39
Completed NSE at 22:39, 0.00s elapsed
Initiating SYN Stealth Scan at 22:39
Scanning localhost (127.0.0.1) [1000 ports]
Discovered open port 22/tcp on 127.0.0.1
Completed SYN Stealth Scan at 22:39, 0.04s elapsed (1000 total ports)
NSE: Script scanning 127.0.0.1.
Initiating NSE at 22:39
Completed NSE at 22:39, 0.09s elapsed
Initiating NSE at 22:39
Completed NSE at 22:39, 0.00s elapsed
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000011s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

NSE: Script Post-scanning.
Initiating NSE at 22:39
Completed NSE at 22:39, 0.00s elapsed
Initiating NSE at 22:39
Completed NSE at 22:39, 0.00s elapsed
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 10.30 seconds
Raw packets sent: 1000 (44.000KB) | Rcvd: 2001 (84.044KB)
```

```
(root@kali)-[/home/kali]
# sudo nmap -sV --script vuln localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-17 00:28 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 8 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:10.0p2:
|     CVE-2025-61985  3.6      https://vulners.com/cve/CVE-2025-61985
|     CVE-2025-61984  3.6      https://vulners.com/cve/CVE-2025-61984
|     B7EACB4F-A5CF-5C5A-809F-E03CCE2AB150  3.6      https://vulners.com/g
|_    ithubexploit/B7EACB4F-A5CF-5C5A-809F-E03CCE2AB150  *EXPLOIT*
|     4C6E2182-0E99-5626-83F6-1646DD648C57  3.6      https://vulners.com/g
|_    ithubexploit/4C6E2182-0E99-5626-83F6-1646DD648C57  *EXPLOIT*
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.06 seconds
```

Interpretation: The first vulnerability scan did not flag any vulnerabilities. After adjusting the command for the second vulnerability scan it did flag 2 CVEs (Common Vulnerabilities and Exposures). The version of OpenSSH installed on the target device is linked to CVE-2025-61984 and CVE-2025-61985 (affect OpenSSH before 10.1; related to ProxyCommand/control-character handling). These CVEs can enable code execution only in specific configurations. Further verification is required.